VITAL AGRI NUTRIENTS

# Sugarcane nutrition: cane yield and sugar recovery.

This deck shows where Punjab's cane crop loses yield and sugar recovery, what the published data say, and what a nutrition programme can and cannot do. It is written for the mill and the grower, from the same evidence.

**2,990,000**
acres of cane in Pakistan, converted from the hectares given by the US Department of Agriculture (USDA)

**700 maunds**
an acre (approximate): Pakistan's 5-year average cane yield, about 28 t (69 t/ha in the USDA report; 1 maund = 40 kg)

**770,160**
Punjab soil samples, tested in about 2016 to 2018

November 2026 · Every figure in this deck is published with its source named, or is marked derived; the arithmetic is on the slide or in the sources on slide 16.

PART 1 · FOR THE MILL

# A mill pays for cane by the tonne, but sugar recovery is decided in the field.

TONNES CRUSHED

## Bought

Tonnage depends on the area planted, the variety and how the season goes for the grower. The mill can contract and pay for it.

SUGAR RECOVERY

## Set in the field

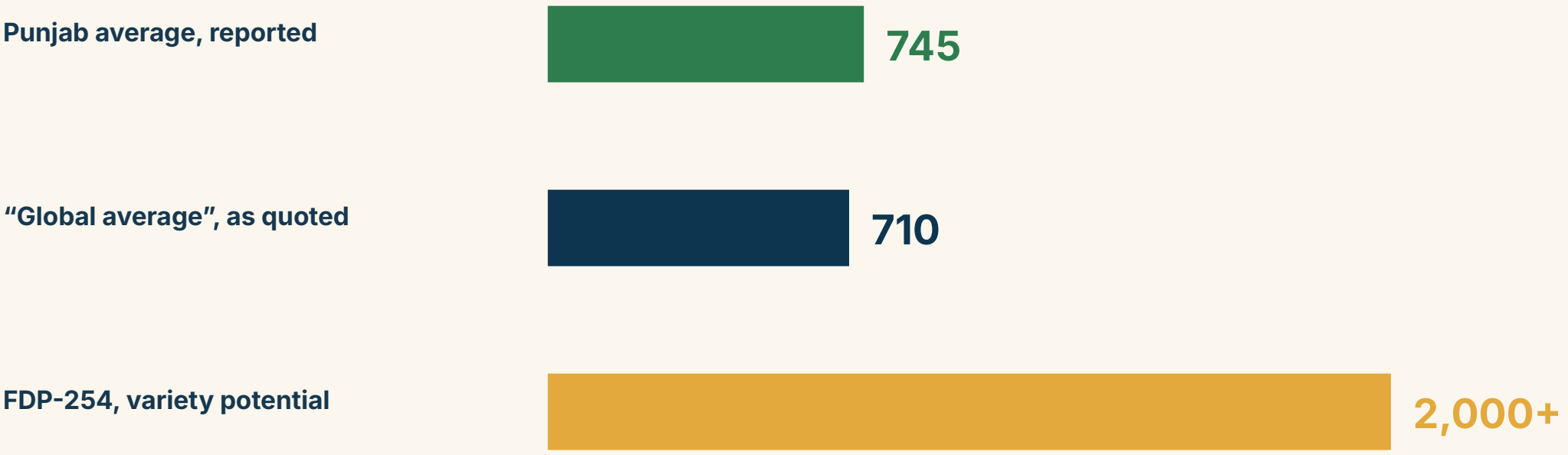
Sugar recovery is determined in the field by the variety, the crop's nutrition during the season and its maturity at harvest.

**Low sugar recovery is an agronomy problem that the mill does not control and does not usually fund. The rest of this deck deals with the part of sugar recovery that nutrition affects.**

This slide describes how sugar recovery is set. VAN publishes no recovery figure for any of its products.

THE STARTING POINT

# Punjab’s average yield is above the quoted global average. FDP-254’s trial potential is about 2.7 times as high.




745 maunds = 29.8 t/acre = 73.6 t/ha. Derived at 1 maund = 40 kg, 1 ha = 2.4710538 acres.

Punjab and global figures: Dr Ishtiaq Hassan, Director General, Adaptive Research and Farm Training Punjab, quoted in the trade press, November 2025. FDP-254 potential and its recovery of up to 12.77%: Dr Muhammad Kashif Munir, Chief Scientist, Sugarcane Research, Ayub Agricultural Research Institute (AARI), in the same trade press report. A variety’s trial potential is not a field average, and this deck does not treat it as one.

WHAT EACH TONNE OF CANE TAKES OUT

# Each tonne of harvested cane takes out 1.8 kg of potash ($K_2O$), by the IPNI coefficient.

**1.8 kg**

potash ($K_2O$) taken out per tonne of cane

**1.0 kg**

nitrogen (N) taken out per tonne of cane

**0.65 kg**

phosphate ($P_2O_5$) taken out per tonne of cane

**A crop of 745 maunds per acre therefore takes out:**

**53.6 kg**

$K_2O$

**29.8 kg**

N

**19.4 kg**

$P_2O_5$

Derived: each coefficient × 29.8 tonnes.

Source · International Plant Nutrition Institute (IPNI) nutrient uptake and removal tables, metric edition 01/15, North America compilation, row “Sugarcane”, per tonne of cane. IPNI’s own note applies: “Reported nutrient removal coefficients may vary regionally depending on growing conditions. Use locally available data whenever possible.” No Pakistani removal table exists on the public record; once one is published, it should replace these coefficients.

WHAT IS APPLIED AGAINST WHAT IS TAKEN OUT

# Against estimated removal, the crop plan applies more N and $P_2O_5$ but 22.3 kg less $K_2O$ per acre.

| Nutrient | VAN crop plan applies | Taken out in the harvested crop | Balance |
|---|---|---|---|
| **Nitrogen (N)** | 71.5 kg | 29.8 kg | **+41.7 kg** |
| **Phosphate ($P_2O_5$)** | 36.4 kg | 19.4 kg | **+17.0 kg** |
| **Potash ($K_2O$)** | 31.3 kg | 53.6 kg | **−22.3 kg** |
| **Sulfur (S)** | 17.1 kg | no coefficient | **n/a** |

No nutrition plan is written to replace everything a harvest removes, so a negative potash balance does not mean the plan is wrong. It shows why the grower should test the soil and not rely only on a published rate.

Applied · VAN's sugarcane (February planting) plant-crop plan as on van.com.pk, per acre, its own rows and printed analyses, multiplied out. The land preparation row follows VAN's updated calculator (Green Phosphate, 2 bags); the published PDF is being reissued. Humi Grow is counted by its printed $K_2O$, 7%. Taken out · IPNI, as on slide 4, at 29.8 t/acre. Derived.

THE SOIL

# Of 770,160 Punjab samples, 87.8% were below the adequate potassium level, and none was tested for sulfur.

**87.8%**

**below the adequate level of potassium**

adequate: 180 ppm · 132 ppm mean

**59.0%**

**low in zinc**

1.25 ppm mean

**51.7%**

**low in boron**

0.59 ppm mean

**90.2%**

**below 0.86% organic matter**

0.61% mean

**96.1%**

**above pH 7.5**

8.22 mean pH

**0**

**samples tested for sulfur**

of 770,160

Source · Punjab Soil Testing Programme district workbooks, 770,160 samples tested in about 2016 to 2018 across 36 districts, each workbook read in full. Every district workbook carries the same 13 determinations, and sulfur is not among them: sulfur is absent from the testing schedule, not missing from the results.

THE MECHANISM

# Potassium is the nutrient that moves sugar out of the leaf and into the stalk.

01

### The leaf makes sucrose

It is made by photosynthesis in the canopy.

02

### Potassium loads the phloem

Potassium drives the transport that carries sucrose out of the leaf, through the phloem: the channels that carry sugar.

03

### Sugar is stored in the stalk

The stalk is what the mill crushes.

04

### Water use is controlled

Potassium controls the stomata, the pores in the leaf, and so affects how the crop copes with heat and water shortage.

**A crop that lacks potassium can still make sugar in the leaf but fail to move it into the stalk. This is the part of sugar recovery that a nutrition programme can affect. VAN publishes no recovery figure, because it has not measured one.**

This slide states standard plant physiology as a mechanism. It claims no yield figure for any VAN product, and none should be inferred from it.

PART 2

# For the grower:
# yield needs every task on time.

The tasks: sowing date, seed and plant population, fertiliser at the right stage, irrigation on schedule, and control of weeds, pests and disease. Crop nutrition is 1 of these tasks. A task missed at its time can cost yield that extra work later may not recover.

NITROGEN USE EFFICIENCY

# In 2023, 29% of the nitrogen added to Pakistan’s cropland was taken out in the harvested crop.

**29%**

Pakistan’s nitrogen use efficiency

**71%**

United States, same measure (approximate)

Part of the loss is caused by soil conditions that the grower does not control. 69% of the Punjab soil samples tested in about 2016 to 2018 were above pH 8. In such soil, part of any urea left on the surface turns to ammonia gas and is lost to the air before the root can take it up. On calcareous soil (soil rich in lime), phosphate also reacts with calcium and slowly becomes unavailable.

01

### Place the nitrogen

Put it in a band: drill it at sowing, or side-dress it and water it in.

02

### Split the nitrogen

Apply it in parts to match the crop’s demand, timed with irrigation.

03

### Use coated urea

Coated urea releases its nitrogen over days, not all at once on a hot soil surface.

04

### Mix sprays correctly

Add products to the tank in the right order, test the mix in a jar first, and spray on the day you mix.

Source · FAOSTAT, Cropland Nutrient Balance, December 2025 release (2023, FAO estimates). Corroborated by Shahzad et al. (2019), Nature Sustainability. Soil figures from the Punjab survey named on slide 6. The 4 steps are standard published practice, not a VAN compatibility chart.

VAN'S SUGARCANE CROP PLAN

# The crop plan has 5 stages and 6 nutrient groups, with the products for each.

| | Land Preparation | Germination | Early Growth | Grand Growth | Maturity |
|---|---|---|---|---|---|
| **Phosphorus** | Green Phosphate $P_2O_5$ 32% · N 6% · V-Phosphate 11-44-0 | | | | |
| **Nitrogen** | Vital Urea N 32% · S 13% | | | | |
| **Potash** | Fusion Potash $K_2O$ 50% · Vital Potash · VL-Potash Liquid | | | | |
| **Secondary** | Green Sulfur S 70% · V-Mag Essential Mg 8.5%, B 1% | | | | |
| **Soil amendment** | V-Compost OM 25% + N 5% · Humi Grow | | | | |
| **Micronutrients** | V-Transform Zinc 21% · VL-Boron 5% | | | | |

The full crop plan, per acre, with every rate and pack size, is at van.com.pk/crops/sugarcane, with the ratoon crop plan beside it.

Source · VAN's published plan, with the land preparation row from VAN's updated calculator (Green Phosphate, 2 bags); the published PDF is being reissued. Every analysis is as printed on the bag. Nothing on this slide is modelled.

THE RATOON CROP

# The ratoon crop starts in soil from which the plant crop has already taken out an estimated 53.6 kg of $K_2O$ per acre.

PLANT CROP, PER ACRE

**71.5 kg**
nitrogen

**31.3 kg**
potash

RATOON, PER ACRE

**87.0 kg**
nitrogen

**31.8 kg**
potash

The ratoon grows from stubble, without a fresh seedbed. Its crop plan applies 22% more nitrogen than the plant-crop plan (87.0 kg against 71.5 kg per acre). Yet the ratoon plan applies about the same potash as the plant crop (31.8 kg against 31.3 kg).

Both columns · VAN's plant-crop and ratoon crop plans as on van.com.pk, per acre, multiplied out; the plant crop's land preparation row follows VAN's updated calculator. Removal figure as on slide 4. Derived.

THE WATER ASSUMPTION

# A nutrition plan assumes that the crop received the water it needed.

Of the crops whose season water the site calculates, plant cane has the longest season: 405 days in FAO’s table. For each district, the sugarcane page on van.com.pk gives the water the crop needs over the season, the water that normal rainfall supplies, and the difference, in acre-inches and in irrigations.

### Limits of the water figure

The reference water use is calculated by the Hargreaves method, not by FAO’s standard equation, which needs humidity, wind and solar radiation; the PMD normals give only temperature and rainfall. In the dry season, Punjab is drier and windier than the moderately humid (sub-humid) climate that FAO’s crop coefficients, the factors that turn reference water use into crop water use, assume. FAO’s own correction for arid climates would raise those coefficients. For both reasons the published water figure is uncertain and probably too low.

### FAO Paper 56

the crop water method of the Food and Agriculture Organization (FAO), used throughout

### PMD normals

Pakistan Meteorological Department (PMD): the temperature and rainfall data used

### Hargreaves

the reference method used, because humidity, wind and radiation are not published for these stations

### Limit stated

the sugarcane page says FAO’s cane row is for low latitudes, the nearest FAO offers for Punjab

FAO Irrigation and Drainage Paper 56. Reference water use (reference evapotranspiration) by Hargreaves on Pakistan Meteorological Department normals. The station used and its distance from the district are named on the page.

WHAT VAN WILL NOT PUBLISH YET

# The cane yield assessment is being built, and it needs a cane weight model.

### WHAT IT WILL DO

A frame is placed in the field, the millable canes inside it are counted, and the yield is read from the count. This measures the crop that is actually standing, unlike a prediction from how the season was farmed. Comparing the measured yield with the predicted yield will give VAN its first real check of the prediction.

### WHAT IS MISSING

A model that gives cane weight from girth and length is missing. VAN's own assessment sheet measures length, girth (as thickness), nodes and node length, but then applies the same weight to every cane. VAN's 2 versions of the sheet use cane weights that differ by about 1/3. Until the link between girth, length and weight is measured, the tool can only count canes; it cannot estimate yield.

**1 season of paired measurements, recording girth, length and weight for the same canes, would give the data the model needs. A mill is better placed than VAN to lead this work, with VAN as a partner.**

ABOUT VAN

# VAN formulates, manufactures, registers and tests its products on 1 site in Pakistan.

**23**

registered brands

**28**

published crop plans, per acre

**21**

laboratory tests, all with published prices

**LAB 336**

accredited by PNAC to ISO/IEC 17025:2017

### What VAN does not claim

VAN gives no yield percentage for a product without a named trial and no recovery figure for sugarcane. It uses no Pakistani removal coefficient, because none is published. Where a number does not exist, VAN says so rather than supplying one, even if a partner asks for one for its marketing.

Every batch is tested before it ships. A farmer can read the certificate of analysis for any bag on van.com.pk by entering the batch number printed on it.

WHAT VAN PROPOSES

# 3 steps, in order of the return they give.

## 01 A soil test across the supply zone

A soil test in the districts that supply cane to the mill, read against the Punjab adequacy levels, so that potassium and zinc levels are measured for the supply zone instead of being taken from the Punjab average.

## 02 A joint measurement for the cane weight model

1 season of paired measurements of girth, length and cane weight. The model would give a yield assessment for every grower the mill contracts with, and VAN would publish the method.

## 03 A programme made for the mill

VAN formulates, manufactures, registers and tests the products, and the bag can carry the mill's own name. There is no minimum order. A production batch is 1 tonne, and a smaller order is packed from a 1 tonne batch.

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SOURCES

# The source of every figure in this deck.

| | |
|---|---|
| **Cane area, production and yield** | USDA Foreign Agricultural Service, Sugar Annual: Pakistan, PK2025-0005. 1,210,000 ha; 80 million tonnes (2024/25, post estimate); 69 t/ha 5-year average. Slide 1 converts the area and the yield to acres at 1 ha = 2.4710538 acres, and the yield to maunds at 1 maund = 40 kg. |
| **Punjab and "global" yield, and FDP-254** | Dr Ishtiaq Hassan, Director General, Adaptive Research and Farm Training Punjab, and Dr Muhammad Kashif Munir, Chief Scientist, AARI Sugarcane Research, both quoted in the trade press, November 2025. These are press quotations, not a dataset. |
| **Nutrient removal per tonne of cane** | IPNI nutrient uptake and removal tables, metric edition 01/15, North America compilation. No Pakistani table exists on the public record. |
| **The soil figures** | Punjab Soil Testing Programme district workbooks: 770,160 samples tested in about 2016 to 2018, 36 districts, 13 determinations, each workbook read in full. |
| **Nitrogen use efficiency** | FAOSTAT, Cropland Nutrient Balance, December 2025 release (2023, FAO estimates). Corroborated by Shahzad et al. (2019), Nature Sustainability. |
| **Water** | FAO Irrigation and Drainage Paper 56, on Pakistan Meteorological Department normals, by the Hargreaves reference method. |
| **The crop plan itself** | VAN's own published sugarcane and ratoon plans, per acre, with the analyses printed on the bags. The plant crop's land preparation row follows VAN's updated calculator (Green Phosphate, 2 bags); that PDF is being reissued. |

**Any figure marked derived is VAN's calculation from the published inputs above; the calculation is on the slide where the figure appears or in the sources above.**